

ELAPSE: Entropy-Linked Autonomous Propagation and Self-Erasure — A Multi-Domain Ensemble Framework for Active Data Containment in Decentralised Networks

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Abstract

We study the statistical physics of information spreading and controlled self-erasure in complex networks. A data payload diffusing on a network drives the spatial entropy $H(t) = -\sum_i p_i \ln p_i$ from a localised initial state ($H \approx 0$) toward maximum entropy ($H = \ln n$), analogous to a thermal relaxation process. We show that this entropy trajectory undergoes a *percolation-like phase transition* at a critical threshold H_c^* : below H_c^* , spontaneous deletion fires before the data infects the bulk (subcritical); above H_c^* , the system reaches maximum exposure before deletion can take effect (supercritical).

We introduce ELAPSE (Entropy-Linked Autonomous Propagation and Self-Erasure), a five-mechanism ensemble that treats deletion as entropy-driven mortality, and establish the following statistical-physical results: (i) the Time-Integrated Entropy Exposure (IEE) $= \int_0^T H(t)M(t) dt$ scales as $\text{IEE} \sim n^\alpha$ with topology-dependent exponent $\alpha \in [1.1, 1.4]$ measured across ER/BA/WS graphs; (ii) the critical entropy threshold satisfies $H_c^* = H_{\max} -$

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$C_0 e^{-2\alpha\lambda_2 T}$, with a transition width that narrows as algebraic connectivity λ_2 increases (narrow for ER; wide for WS); (iii) exact convexity of the linearised IEE landscape, while the full nonlinear surface has a 14.1% local non-convexity rate.

Statistical evaluation over $N_{\text{test}} = 20$ out-of-sample seeds across ER, BA, and WS topologies ($n \in \{50, 100, 200, 500\}$) confirms $\geq 9\times$ IEE reduction over the uncontrolled baseline, with the ensemble achieving lowest topology-averaged IEE at all scales.

Keywords: Complex networks, Information diffusion, Shannon entropy, Phase transition, Percolation, Stochastic differential equations, Spreading phenomena, Entropy-driven deletion, Statistical physics

1. Introduction

1.1. Motivation and Problem Statement

The proliferation of sensitive data across decentralised peer-to-peer (P2P) networks constitutes a qualitatively different privacy threat than the storage of data in centralised repositories. Once a data payload escapes its point of origin — whether via legitimate forwarding, an insider disclosure, or end-point compromise — the originating system loses all physical control over its further replication. The challenge is acute in the context of the European Union’s *right to erasure* (GDPR Article 17 [1]), which mandates that data subjects can demand removal of their personal data; a mandate that is straightforwardly unenforceable once a payload has diffused across thousands of pseudonymous P2P nodes.

The scale of the problem is growing. IBM’s 2024 Cost of a Data Breach

Report [2] documented a global average breach cost of USD 4.88 million, a 10% year-on-year increase, with P2P exfiltration cited among the leading attack vectors. The 2024 Verizon Data Breach Investigations Report [3] further notes that data discovered in unauthorised copies on external networks has a median “time to identification” exceeding 280 days, during which the privacy exposure accumulates continuously.

1.2. *Limitations of Chronological Containment*

The existing active-containment literature — most influentially the Vanish [4] and FADE [5] systems — relies on a shared architectural assumption: a *chronological decay timer* governs data lifetime. Vanish overwrites Distributed Hash Table (DHT) shards on a fixed schedule (typically 8 hours), so that the decryption key is irretrievably lost after a wall-clock timeout. FADE extends this with file-assured deletion via policy-based key management.

The fundamental limitation of this paradigm is its independence from the actual network topology. On a scale-free (Barabási-Albert) graph, the hub-driven spreading mechanism can saturate a majority of nodes within a small fraction of any fixed timeout [6, 7]. A data payload that has already reached 10^3 nodes before the 8-hour timer fires may be practically irrecoverable regardless of key deletion. Conversely, on a sparse small-world (Watts-Strogatz) graph, the same timer may fire when the data has barely left the source neighbourhood, over-triggering deletion.

1.3. *The Entropy-Clock Approach*

We propose that data lifetime should be a *dynamic function of the network state*. Specifically, ELAPSE uses the Shannon entropy of the data’s

spatial distribution as its deletion clock:

$$H(t) = - \sum_{i=1}^n p_i(t) \ln p_i(t), \quad p_i(t) = \frac{x_i(t)}{\sum_j x_j(t)}, \quad (1)$$

where $x_i(t) \geq 0$ is the data concentration at node i at time t . When $H(t)$ is low, data is concentrated on few nodes (low privacy exposure); when $H(t)$ approaches $H_{\max} = \ln n$, data has spread uniformly (maximal exposure). Self-erasure is triggered when $H(t)$ crosses a critical threshold H_c , replacing the chronological clock with a *spatial-topological clock*.

1.4. Contributions

This paper makes the following contributions:

1. **Framework:** We introduce ELAPSE, fusing five mechanism-specific deletion votes via an entropy-regularised Nelder-Mead ensemble optimising IEE (Section 3–4).
2. **Rigorous theory** (Section 4.3): Five results: (i) pathwise-unique non-negative strong solution (Theorem 1, via Yamada–Watanabe); (ii) a closed-form spectral IEE bound in terms of n , M_0 , and deletion intensity (Theorem 2); (iii) a *topology-refined* IEE bound incorporating the algebraic connectivity λ_2 , giving the spectral trigger time $\tau(\lambda_2)$ explicitly (Corollary 1); (iv) a residual-mass privacy guarantee connecting IEE to an exponential data-recovery bound for bounded adversaries (Proposition 1); (v) *exact* convexity of the linearised IEE objective (Theorem 3, Corollary 2). The full nonlinear SDE has a 14.1% Jensen-violation rate, quantifying the linearisation gap.

3. **Statistical rigour:** All IEE values are reported as mean $\pm$ 95% CI over $N_{\text{test}} = 20$ out-of-sample seeds, three topologies, and four network sizes (Section 5).
4. **Real-world validation:** The full ensemble is evaluated on subgraphs of the Stanford SNAP Gnutella P2P datasets [8] (Section 6).
5. **Adversarial analysis:** We model an attacker who throttles propagation and propose an adaptive- H_c countermeasure (Section 7).
6. **Distributed estimation:** We propose a gossip-protocol approximation to the global entropy observable and quantify its trigger-accuracy (Section Appendix A).

1.5. Paper Organisation

Section 2 reviews the literature and positions ELAPSE. Section 3 defines the five core mechanisms. Section 4 presents the ensemble construction and five theoretical results (Theorems 1–3, Corollaries 1–2, Propositions 1–2). Section 5 provides the main statistical evaluation. Section 6 reports real-world validation. Section 7 analyses adversarial evasion. Section Appendix C concludes. Appendix Appendix A covers distributed entropy estimation (gossip protocol); Appendix Appendix B provides a mechanistic analysis and fix for M3’s underperformance.

2. Related Work

2.1. Information Diffusion and Entropy on Graphs

The quantitative study of information propagation on complex networks has a rich lineage. Foundational epidemiological models for spreading processes [7] established that network topology — and especially heterogeneous

degree distributions — fundamentally alters threshold phenomena. The particular role of Shannon entropy as a topological indicator gained prominence through Zhao et al. [9], who demonstrated that entropy-spreading metrics provide accurate macroscopic proxies for information volatility across heterogeneous degree distributions, establishing $H(t)$ as a reliable outbreak-detection signal.

Parallel advances in continuous graph diffusion, originating in spectral graph signal processing [10] and extended through neural PDE-on-graph formulations (GRAND [11]; Eliasof et al. [12]; Fan et al. [13]), confirm that partial differential equations on graph topologies accurately capture non-linear propagation boundaries. Universal resilience patterns [14] further characterise scale-stress behaviour under perturbation.

2.2. Static and Active Data Containment

Early cryptographic containment schemes were pioneered by Vanish [4], which exploited DHT key churn to implement passive key decay. Wolchok et al. [15] demonstrated Sybil-attack vulnerability. FADE [5] and the Ephemerizer [16] strengthened storage layers but retained chronological timers. Rondeau and Temple [17] survey secure deletion approaches and confirm that time-blind mechanisms are structurally insufficient for dynamic network environments. Truex et al. [18] demonstrate a hybrid local/global approach to privacy-preserving federated learning in distributed settings, showing that no single privacy mechanism dominates across heterogeneous network topologies — a finding consistent with our ensemble design.

2.3. Machine Unlearning and Algorithmic Forgetting

The machine-learning community has recently formalised the complementary problem of “algorithmic forgetting”: ensuring a trained model provides no residual information about a removed training sample. Bourtole et al. [19] established the SISA training framework, Nguyen et al. [20] and Xu et al. [21] provide comprehensive surveys of unlearning algorithms, highlighting the need for topological removal verification. Oesterling et al. [22] incorporate fairness constraints into active data removal, which motivates our entropy-regularised ensemble approach.

2.4. Differential Privacy and Information-Theoretic Bounds

Dwork and Roth’s foundational treatise [23] establishes differential privacy as the canonical framework for information-theoretic privacy guarantees. While differential privacy bounds *inference leakage* from a fixed dataset, ELAPSE addresses a complementary problem: minimising the *spatial exposure* of a diffusing payload before deletion can take effect. These two regimes are formally distinct but share the use of entropy-based metrics as core sensitivity measures.

2.5. Control of Diffusion Processes on Graphs

ELAPSE is related to, but distinct from, the emerging literature on *controlled diffusion on networks*. Mean-field game (MFG) frameworks [24, 25] study populations of McKean–Vlasov-type agents interacting through their empirical distribution, deriving feedback control laws from coupled Hamilton–Jacobi–Bellman and Fokker–Planck equations. In MFG language, ELAPSE is a *social planner* problem: we seek a mortality schedule

$\mu\Delta(t; \mathbf{w})$ that minimises a population-level cost (IEE) subject to the coupled McKean–Vlasov SDE (??). However, ELAPSE differs from classical MFG in three ways: (i) topology enters via the Laplacian L rather than a mean-field coupling term; (ii) the control $\Delta(t; \mathbf{w})$ is an entropy-dependent functional of the empirical distribution; and (iii) we optimise over a finite ensemble of pre-specified mechanisms rather than solving a PDE. Online feedback control on network flow manifolds [26] addresses a related setting but focuses on power systems rather than information-theoretic objectives. ELAPSE contributes to this space with the first entropy-functional IEE bound (Theorem 2) and spectral trigger time (Corollary 1) for controlled diffusion on heterogeneous P2P graphs.

2.6. Positioning of ELAPSE

Table 1 summarises how ELAPSE differs from the three dominant paradigms for data-lifetime management.

ELAPSE occupies a unique position in the landscape: it is the first framework that (i) replaces the chronological erasure clock with a spatial entropy trigger, (ii) constructs a regularised ensemble across five physical domains, (iii) provides rigorous theoretical guarantees (Theorems 1–3) backed by statistical evaluation, and (iv) characterises the minimum gossip hop count k^* as a function of network algebraic connectivity (Proposition 3). Prior ensemble approaches for network spreading (e.g., competing epidemics [27]; social cascades [28]) do not address active containment.

Table 1: Comparison of data-deletion paradigms. ELAPSE is the only approach that is simultaneously topology-aware, operates at network level, minimises a formal IEE objective, and provides distributed estimation.

Paradigm	Topology- aware	Network- level	Formal IEE obj.	Distributed estimation
Cryptographic (Vanish, FADE) [4, 5]	✗	✗	✗	✗
Chron. timer (M7, §5.8)	✗	✗	✗	✗
Machine unlearning [19, 21]	✗	✗	✗	✗
Differential privacy [23]	✗	✗	✗	Partial
ELAPSE (ours)	✓	✓	✓	✓

3. The Five Core Mechanisms

3.1. Network and Notation

Let $G = (V, E)$ be an undirected connected graph with $|V| = n$ nodes. The graph Laplacian $L = D - A$ (where D is the degree matrix and A the adjacency matrix) and the Fiedler value λ_2 (second-smallest eigenvalue of L , the algebraic connectivity [29]) play central roles. The data concentration at node i at time t is $x_i(t) \geq 0$, with units of *data mass per node*. Initial conditions are $x_i(0) = u_i$ where $u_i \sim \text{Uniform}(0.5, 1.0)$ for a randomly chosen 10% of nodes (the initial data sources) and $x_i(0) = 0$ otherwise; total initial mass $M(0) = \sum_i x_i(0) \in (0, n/10]$. All IEE values therefore have units of $[\text{nats} \times \text{data-mass} \times \text{time}]$, and scale with both n and $M(0)$. Spatial privacy exposure is measured by $H(t)$ as defined in Eq. (1), with $H_{\max} = \ln n$ and a critical threshold $H_c = 0.65H_{\max}$ used throughout.

Each of the five mechanisms defines a continuous *deletion vote* $v_k(t) \in [0, 1]$, measuring urgency of erasure. The composite mortality pressure $\Delta(t) = \sum_k w_k v_k(t)$ drives data removal.

3.2. M1: Entropy-Gated Diffusion Model (EGDM)

The physical baseline treats data as a field propagating against a graph-Laplacian operator with an entropy-triggered global mortality term:

$$\frac{d\mathbf{x}}{dt} = -\alpha L\mathbf{x} - \mu \sigma(H(t)) \mathbf{x} + \mathbf{s}(t) + dW_t, \quad (2)$$

where $\alpha = 0.3$ is the diffusion coefficient, $\mu = 1.5$ the mortality rate, $\mathbf{s}(t)$ a small external injection, and dW_t a multiplicative CIR-type Wiener noise with scale $\sigma_n = 0.015$. The polynomial vote is:

$$v_1(t) = \left[\max\left(0, \frac{H(t) - H_c}{H_{\max} - H_c}\right) \right]^\beta, \quad (3)$$

with sharpness $\beta = 2$. Near-zero below H_c , the vote rises as a smooth power-law as data spreads. Strong Lipschitz continuity of the deterministic flow ensures existence and uniqueness of strong solutions up to the erasure threshold [30].

3.3. M2: Epidemic-Entropy (Compartmental SIR)

Epidemiology models data spread as an infectious load. Nodes transition through Susceptible-Infected-Recovered compartments, capturing local neighbour-to-neighbour spreading independently of global diffusion:

$$\frac{dI_i}{dt} = \beta_{\text{sir}} S_i \sum_j A_{ij} I_j - \gamma I_i - \mu \sigma(H(t)) I_i, \quad (4)$$

where $\beta_{\text{sir}} = 0.4$, $\gamma = 0.15$. The vote is the current infected fraction: $v_2(t) = n^{-1} \sum_i I_i(t)$. This mechanism fires rapidly on dense-core topologies but is suppressed by recovery on sparse graphs.

3.4. M3: Stochastic-Finance (Ornstein-Uhlenbeck First-Passage)

Drawing from barrier-option pricing [31, 32], information exposure risk is modelled as a mean-reverting stochastic process approaching a ruin boundary. The state evolves as:

$$d\mathbf{x} = \theta(\mu_{ou} - \mathbf{x}) dt + \sigma_{ou} dW_t - \mu P(\tau \leq t) \mathbf{x} dt, \quad (5)$$

with $\theta = 0.3$, $\mu_{ou} = 0.1$, $\sigma_{ou} = 0.04$. The vote is the first-passage probability:

$$v_3(t) = 1 - \exp(-\lambda_{fp} t), \quad \lambda_{fp} = \frac{\alpha \lambda_2}{H_{\max} - H_c}, \quad (6)$$

encoding the spectral geometry of the network through the Fiedler value λ_2 . As analysed in Section Appendix B, the static use of λ_2 limits M3's responsiveness to fast BA hub dynamics.

3.5. M4: Biological Cooperative Binding (Hill Function)

In systems biology, cooperative binding transitions are modelled by the Hill function [33]:

$$v_4(t) = \frac{H(t)^{n_h}}{H_c^{n_h} + H(t)^{n_h}}, \quad (7)$$

with Hill coefficient $n_h = 4$. This produces a smoother, more Lyapunov-stable asymptotic transition than the polynomial v_1 : it is near zero for $H \ll H_c$, exactly $\frac{1}{2}$ at $H = H_c$, and near one for $H \gg H_c$. The sigmoidal shape is mathematically well-characterised from gene regulatory network analysis.

3.6. M5: Sociological Cascade Pressure

Sociological cascade models parameterise agents that delete based on localised sub-graph density [34, 28].

Definition 1 (Indicator notation). *For any predicate P , let $\mathbf{1}[P]$ denote the indicator function: $\mathbf{1}[P] = 1$ if P is true, $\mathbf{1}[P] = 0$ otherwise. Specifically, for a threshold κ and deletion threshold ϵ : $\mathbf{1}[x_j < \epsilon]$ equals 1 if node j has effectively deleted its payload (concentration below $\epsilon = 0.05$), and 0 otherwise; $\mathbf{1}[\cdot > \kappa]$ equals 1 if the argument exceeds the cascade threshold $\kappa = 0.3$.*

The cascade vote combines global and local deletion pressure:

$$v_5(t) = \frac{1}{2}\sigma(H(t)) + \frac{1}{2n} \sum_{i=1}^n \mathbf{1} \left[\frac{\sum_{j \in N(i)} \mathbf{1}[x_j < \epsilon]}{\deg(i)} > \kappa \right], \quad (8)$$

where $N(i)$ is the neighbourhood of node i and $\deg(i)$ its degree. A node responds to deletion when the global entropy exceeds H_c or when more than κ of its neighbours have already deleted.

4. The ELAPSE Ensemble

4.1. Ensemble Architecture

The M6 ELAPSE engine fuses the five votes into a terminal deletion pressure:

$$\Delta(t) = \mathbf{w}^\top \mathbf{v}(t), \quad \mathbf{w} \in \mathcal{W} = \{\mathbf{w} \in \mathbb{R}_{\geq 0}^5 : \|\mathbf{w}\|_1 = 1\}, \quad (9)$$

where $\mathbf{v}(t) = [v_1(t), \dots, v_5(t)]^\top$. The Time-Integrated Entropy Exposure (IEE) is:

$$\text{IEE}(\mathbf{w}) = \int_0^T H(\mathbf{p}(t; \mathbf{w})) \cdot M(t; \mathbf{w}) dt, \quad (10)$$

where $M(t; \mathbf{w}) = \sum_i x_i(t)$ is the surviving data mass. A lower IEE indicates better privacy protection.

4.2. Entropy-Regularised Optimisation

Unconstrained Nelder-Mead on $\mathcal{W}$ collapses to a one-hot vertex $\mathbf{e}_k$ (choosing the single best mechanism), sacrificing generalisation across topology types. We add an entropy regulariser:

$$\min_{\mathbf{w} \in \mathcal{W}} J(\mathbf{w}) = \text{IEE}(\mathbf{w}) - \lambda_{\text{reg}} \mathcal{H}(\mathbf{w}), \quad (11)$$

where $\mathcal{H}(\mathbf{w}) = -\sum_k w_k \ln w_k$ is the weight-layer entropy. Subtracting this term penalises low-entropy (collapsed) weight distributions while allowing IEE minimisation to proceed. We set $\lambda_{\text{reg}} = 15$ throughout, selected at the Pareto elbow between IEE and weight diversity (see Table 5).

4.3. Theoretical Analysis

We establish three main results: well-posedness of the SDE system, a closed-form spectral upper bound on IEE, and rigorous convexity of the linearised IEE objective.

Theorem 1 (Well-posedness of ELAPSE). *Let G be connected and $\mathbf{x}(0) \geq \mathbf{0}$ componentwise. The ELAPSE SDE system (2)–(9) has a pathwise-unique, non-negative strong solution $\mathbf{x}(t) \geq \mathbf{0}$ a.s. on $[0, T]$ for any finite $T > 0$.*

Proof. We establish the four required properties in sequence.

(i) Local existence via locally Lipschitz drift. Each vote $v_k(t; \mathbf{x}) \in [0, 1]$ is a continuous function of $H(\mathbf{x})$, and H is C^1 on the open set $\Omega_c = \{\mathbf{x} : M(\mathbf{x}) \geq c > 0\}$. On any compact $K \subset \Omega_c$, the drift $b_i(\mathbf{x}, t) = -\alpha(L\mathbf{x})_i - \mu\Delta(t; \mathbf{w})x_i + s_i(t)$ is therefore Lipschitz in $\mathbf{x}$ with constant $L_K = \alpha\|L\|_2 + \mu + \|\nabla_{\mathbf{x}}H\|_{C(K)}$. Local existence and uniqueness on $[0, \tau_K)$ (up to the first exit from K) follows from the Picard–Lindelöf theorem for SDEs [30].

(ii) **Global existence via Lyapunov bound.** Let $V(\mathbf{x}) = M(\mathbf{x}) = \sum_i x_i$. Itô's formula gives:

$$dM = \left(-\mu\Delta(t; \mathbf{w})M + \|\mathbf{s}\|_1 \right) dt + \sigma_n \sum_i \sqrt{x_i} dW_i,$$

using $\mathbf{1}^\top L = \mathbf{0}$. Taking expectations: $d\mathbb{E}[M]/dt \leq \|\mathbf{s}\|_1$, so $\mathbb{E}[M(t)] \leq M_0 + \|\mathbf{s}\|_1 T < \infty$ for all $t \in [0, T]$. By Markov's inequality the solution does not exit Ω_c in finite expected time, ruling out blow-up and allowing extension to $[0, T]$.

(iii) **Non-negativity.** Fix any component i . When $x_i = 0$ the diffusion coefficient $\sigma_n \sqrt{x_i}$ vanishes exactly, and the drift reduces to $b_i|_{x_i=0} = \alpha \sum_j A_{ij} x_j + s_i \geq 0$ (inward Laplacian flux plus source injection). The component x_i therefore satisfies the CIR-type SDE $dx_i \geq (\alpha \sum_j A_{ij} x_j + s_i) dt + \sigma_n \sqrt{x_i} dW_i$ at the boundary. By the comparison theorem for CIR processes (Feller [35]; see also Theorem IX.3.7 in [36]), $Y \equiv 0$ is a strict lower barrier: $x_i(t) \geq 0$ a.s. on $[0, T]$.

(iv) **Pathwise uniqueness via Yamada–Watanabe.** The diffusion coefficient $\sigma(u) = \sigma_n \sqrt{u}$ satisfies $|\sigma(u) - \sigma(v)| = \sigma_n |\sqrt{u} - \sqrt{v}|$. For $u, v \geq 0$: $(\sqrt{u} - \sqrt{v})^2(\sqrt{u} + \sqrt{v})^2 = (u - v)^2$, so $|\sqrt{u} - \sqrt{v}| = |u - v|/(\sqrt{u} + \sqrt{v}) \leq \sqrt{|u - v|}$, giving $|\sigma(u) - \sigma(v)| \leq \sigma_n \sqrt{|u - v|}$. With $h(r) = \sigma_n \sqrt{r}$: $\int_0^1 h(r)^{-2} dr = \sigma_n^{-2} \int_0^1 r^{-1} dr = +\infty$, verifying the Yamada–Watanabe Hölder- $\frac{1}{2}$ condition (Theorem IX.3.5 in [36]). Combined with (i)–(iii), the Yamada–Watanabe theorem yields a pathwise-unique strong solution. $\square$

Lemma 1 (Positivity of deletion intensity). *Let $\mathbf{w}^* \in \text{int}(\mathcal{W})$ be the ensemble weight vector guaranteed by Corollary 2. Define $\tau^* = \tau(\lambda_2)$ from (15). Then:*

$$\delta_{\min} := \inf_{t \in [\tau^*, T]} \Delta(t; \mathbf{w}^*) > 0. \quad (12)$$

Proof. Corollary 1 gives $H(t) \geq H_c$ for all $t \geq \tau^*$. Since $H(t) > H_c$ on a set of positive measure (strict inequality holds except at the measure-zero boundary), the M1 vote satisfies $v_1(t) = [(H(t) - H_c)/(H_{\max} - H_c)]^\beta > 0$ on $(\tau^*, T]$. As $\mathbf{w}^* \in \text{int}(\mathcal{W})$ we have $w_1^* > 0$, so $\Delta(t; \mathbf{w}^*) \geq w_1^* v_1(t) > 0$. Since v_1 is continuous and strictly positive on the compact set $[\tau^*, T]$, its infimum is strictly positive; hence $\delta_{\min} > 0$. $\square$

Theorem 2 (Spectral IEE bound). *Let $\delta_{\min} > 0$ be as in Lemma 1 (positivity guaranteed by Corollary 2). Let $\tau^* = \tau(\lambda_2)$ from (15) be the spectral trigger time. The IEE of the controlled system satisfies:*

- (a) Pre-trigger phase $[0, \tau^*]$: $\int_0^{\tau^*} H(t)M(t) dt \leq \ln(n) M_0 \tau^*$.
- (b) Post-trigger phase $[\tau^*, T]$: For the deterministic system ($\sigma_n = 0$, $\mathbf{s} \equiv \mathbf{0}$):

$$\int_{\tau^*}^T H(t)M(t) dt \leq \ln(n) \cdot M_0 \cdot \frac{1 - e^{-\mu\delta_{\min}(T-\tau^*)}}{\mu\delta_{\min}}. \quad (13)$$

Combining (a) and (b):

$$\text{IEE}(\mathbf{w}) \leq \ln(n) M_0 \left[\tau^* + \frac{1 - e^{-\mu\delta_{\min}(T-\tau^*)}}{\mu\delta_{\min}} \right]. \quad (14)$$

Proof. Part (a): Since $H(t) \leq \ln n$ and $M(t) \leq M_0$ trivially: $\int_0^{\tau^*} HM dt \leq \ln(n) M_0 \tau^*$.

Part (b): On $[\tau^*, T]$, Lemma 1 gives $\Delta(t; \mathbf{w}^*) \geq \delta_{\min} > 0$. Since $\mathbf{1}^\top L = \mathbf{0}$:

$$\dot{M}(t) = -\mu\Delta(t)M(t) \leq -\mu\delta_{\min}M(t).$$

Grönwall's inequality gives $M(t) \leq M(\tau^*)e^{-\mu\delta_{\min}(t-\tau^*)} \leq M_0e^{-\mu\delta_{\min}(t-\tau^*)}$. Integrating $H(t) \leq \ln n$ over $[\tau^*, T]$ yields (13). Summing parts (a) and (b) gives (14). $\square$

Corollary 1 (Topology-refined IEE bound via algebraic connectivity). *Under the same conditions as Theorem 2, let α be the diffusion coefficient (M1), $\lambda_2 = \lambda_2(L)$ the algebraic connectivity, and $C_0 = n \|p(0) - \mathbf{1}/n\|_2^2$ the normalised initial deviation from uniformity, where $p_i(0) = x_i(0)/M_0$. Define the spectral trigger time:*

$$\tau(\lambda_2) = \frac{1}{2\alpha\lambda_2} \ln \frac{C_0}{H_{\max} - H_c}. \quad (15)$$

Provided $C_0 > H_{\max} - H_c$ (which holds whenever $H(0) < H_c$, guaranteed by our initialisation), the controlled IEE satisfies the tighter bound:

$$\text{IEE}(\mathbf{w}) \leq \ln(n) M_0 \left[\tau(\lambda_2) + \frac{1 - e^{-\mu\delta_{\min}(T - \tau(\lambda_2))}}{\mu\delta_{\min}} \right]. \quad (16)$$

For dense graphs ($\lambda_2 \gg 1$), $\tau \rightarrow 0$ and (16) recovers (13); for sparse graphs ($\lambda_2 \rightarrow 0$), τ grows as $1/(2\alpha\lambda_2)$, revealing the topology-dependent cost of delayed triggering.

Proof. Since the mortality term $-\mu\Delta(t)x_i$ is proportional to x_i , the normalised distribution $p_i(t) = x_i(t)/M(t)$ satisfies $\dot{p} = -\alpha Lp$ (the mortality contributions cancel between numerator and denominator). Spectral decomposition of $e^{-\alpha L t}$ gives:

$$\|p(t) - \mathbf{1}/n\|_2^2 \leq \|p(0) - \mathbf{1}/n\|_2^2 \cdot e^{-2\alpha\lambda_2 t},$$

since all non-constant eigenmodes decay at rate $\geq \lambda_2$. The χ^2 -KL inequality $\text{KL}(p \parallel \mathbf{1}/n) \leq \chi^2(p \parallel \mathbf{1}/n) = n \|p - \mathbf{1}/n\|_2^2$ then yields:

$$H_{\max} - H(t) = \text{KL}\left(p(t) \parallel \frac{\mathbf{1}}{n}\right) \leq C_0 e^{-2\alpha\lambda_2 t}. \quad (17)$$

At $t = \tau(\lambda_2)$ the right-hand side of (17) equals $H_{\max} - H_c$, so $H(t) \geq H_c$ for all $t \geq \tau(\lambda_2)$: the entropy-triggered controller is guaranteed to be active by τ . Bounding $H(t) \leq \ln(n)$ on $[0, \tau]$ and applying Theorem 2's decay argument on $[\tau, T]$ yields (16). $\square$

4.4. Critical Threshold and Phase Transition Near H_c

The entropy threshold H_c plays the role of an *order parameter* for the erasure process. Corollary 1 reveals a percolation-like phase transition:

Remark 1 (Percolation-like transition at H_c). *The spectral trigger time $\tau(\lambda_2)$ in (15) diverges as $H_c \rightarrow H_{\max}$ (supercritical: controller triggers too late, data infects the bulk) and collapses to zero as $H_c \rightarrow H(0)$ (subcritical: premature triggering before meaningful spread). The critical threshold*

$$H_c^* = H_{\max} - C_0 e^{-2\alpha\lambda_2 T} \quad (18)$$

separates these phases. The transition width $\Delta H_c \sim 1/(2\alpha\lambda_2 T)$ narrows as λ_2 increases: on ER ($\lambda_2 \approx 1.8$) the transition is sharp (width ≈ 0.09 in units of $H_{\max}$); on WS ($\lambda_2 \approx 0.02$) the transition is wide (width ≈ 8.3 , comparable to $H_{\max}$ itself), consistent with the gradual sensitivity seen in Figure 5. The phase diagram parametrised by (H_c, λ_2) is analogous to the bond-percolation phase diagram on random graphs [37], with H_c^ playing the role of the percolation threshold p_c .*

Remark 2 (Quantitative topology ordering). *At $n = 100$ with $\alpha = 0.3$ and our initialisation ($C_0 \approx 0.9$, $H_{\max} - H_c \approx 0.35 \ln 100 \approx 1.61$): since $C_0 < H_{\max} - H_c$ the trigger is guaranteed only when ER ($\lambda_2 \approx 1.8$, $\tau \approx 0.48$) and BA ($\lambda_2 \approx 0.04$, $\tau \approx 21.4$) are treated separately. On WS ($\lambda_2 \approx 0.02$,*

$\tau \rightarrow \infty$), the spectral trigger bound is vacuous and the controller relies on the direct IEE bound (13). This ordering — ER triggers fastest, WS slowest — is consistent with the empirical IEE values in Table 3.

Theorem 3 (Linearised IEE is convex in ensemble weights). *Let $\bar{v}_k = T^{-1} \int_0^T v_k(t) dt$ be the time-averaged vote of mechanism k , and $\bar{\Delta}(\mathbf{w}) = \mathbf{w}^\top \bar{\mathbf{v}}$. Under the linearisation that replaces the time-varying deletion rate with its temporal mean (i.e., $M(t; \mathbf{w}) \approx M_0 e^{-\mu \bar{\Delta}(\mathbf{w})t}$ and the spatial distribution is approximately independent of $\mathbf{w}$):*

$$\text{IEE}_{\text{lin}}(\mathbf{w}) = M_0 \int_0^T H_0(t) e^{-\mu \bar{\Delta}(\mathbf{w})t} dt = M_0 \mathcal{L}_{[0,T]} \{H_0\}(\mu \bar{\Delta}(\mathbf{w})), \quad (19)$$

where $H_0(t)$ is the $M0$ entropy trajectory and $\mathcal{L}_{[0,T]}$ denotes the truncated Laplace transform. $\text{IEE}_{\text{lin}}(\mathbf{w})$ is **convex** in $\mathbf{w}$ on $\mathcal{W}$.

Proof. Let $s = \mu \bar{\Delta}(\mathbf{w})$. The second derivative of the Laplace transform with respect to s is:

$$\frac{d^2}{ds^2} \mathcal{L}_{[0,T]} \{H_0\}(s) = \int_0^T H_0(t) t^2 e^{-st} dt \geq 0, \quad (20)$$

since $H_0(t) \geq 0$, $t^2 \geq 0$, $e^{-st} > 0$. Hence $\mathcal{L}_{[0,T]} \{H_0\}$ is convex in s . Since $s = \mu \bar{\Delta}(\mathbf{w})$ is *affine* in $\mathbf{w}$, and the composition of a convex function with an affine map is convex, $\text{IEE}_{\text{lin}}(\mathbf{w})$ is convex in $\mathbf{w}$ on $\mathcal{W}$. $\square$

Corollary 2 (Unique regularised minimum). *For any $\lambda_{\text{reg}} > 0$, the regularised linearised objective $J_{\text{lin}}(\mathbf{w}) = \text{IEE}_{\text{lin}}(\mathbf{w}) - \lambda_{\text{reg}} \mathcal{H}(\mathbf{w})$ has a unique global minimum in $\text{int}(\mathcal{W})$.*

Proof. IEE_{lin} is convex by Theorem 3. The weight entropy $-\mathcal{H}(\mathbf{w})$ is *strictly* convex on $\text{int}(\mathcal{W})$, so J_{lin} is strictly convex. A strictly convex function on

a compact convex set attains a unique global minimum. Since $J_{\text{lin}} \rightarrow +\infty$ as any $w_k \rightarrow 0$ (due to $-\lambda_{\text{reg}} w_k \ln w_k \rightarrow 0$ but the gradient $\rightarrow -\infty$), the minimiser lies in $\text{int}(\mathcal{W})$. $\square$

Remark 3 (Scope of Corollary 2). *Corollary 2 establishes uniqueness for the linearised surrogate lin ; the full nonlinear objective $\text{IEE}(\mathbf{w})$ is optimised empirically via multi-restart Nelder-Mead (Section 4.5). The 14.1% Jensen-violation rate (Section 4.5) quantifies the gap: $\mathbf{w}_{\text{lin}}^*$ is guaranteed unique and serves as a warm start; $\mathbf{w}_{\text{NM}}^*$ coincides with it in the 85.9% of weight space where the linearisation is accurate.*

Proposition 1 (IEE bound implies residual-mass privacy guarantee). *From Theorem 2, the surviving data mass at time T satisfies $M(T) \leq M_0 \varepsilon(T)$ where $\varepsilon(T) = e^{-\mu \delta_{\min} T}$. An adversary who can inspect Q uniformly random nodes at time T recovers expected data mass:*

$$\mathbb{E}[\text{recovered mass}] \leq \frac{Q}{n} \cdot M_0 \varepsilon(T). \quad (21)$$

The adversary's expected recovery fraction $Q\varepsilon(T)/n$ is equivalent to Q -fold $\varepsilon(T)$ -approximate data indistinguishability: choosing $\mu \delta_{\min} \geq T^{-1} \ln(Qn/\eta)$ ensures recovery probability at most η/n per node — no better than blind random search.

Proof. Homogeneous mortality reduces $M(t)$ at rate $\mu \Delta(t) \geq \mu \delta_{\min}$, so $\mathbb{E}[M(T)] \leq M_0 e^{-\mu \delta_{\min} T} = M_0 \varepsilon(T)$. Each queried node holds expected mass $\mathbb{E}[x_i(T)] = \mathbb{E}[M(T)]/n$ by symmetry (the spatial distribution is exchangeable in expectation over random graph realisations). Linearity of expectation over Q queries gives (21). Setting $Q\varepsilon(T)/n \leq \eta/n$ and solving yields the stated condition on $\mu \delta_{\min}$. $\square$

Remark 4 (Relation to differential privacy). *Equation (21) provides a mass-retention privacy guarantee complementary to differential privacy [23]: DP bounds inference leakage from a fixed dataset, whereas Proposition 1 bounds the physical recovery of data whose network lifetime is controlled by ELAPSE. The two frameworks are formally distinct but both encode privacy via an exponential decay parameter (ε in DP; $\varepsilon(T) = e^{-\mu\delta_{\min}T}$ here).*

Empirical Conjecture 1 (Approximate ensemble dominance). This statement is empirically supported but is not analytically established for the full nonlinear SDE; it holds exactly for the linearised system (Theorems 3–2). The nonlinear IEE surface exhibits a 14.1% Jensen-violation rate (Section 4.5), so the claim below is an empirical conjecture. Formal proof for the nonlinear case remains open.

Let $\mathbf{w}^ = \arg \min_{\mathbf{w} \in \mathcal{W}} J(\mathbf{w})$ be the output of multi-restart Nelder-Mead on the full nonlinear objective. By Corollary 2, the linearised optimal weight $\mathbf{w}_{\text{lin}}^*$ is unique and interior, ensuring the ensemble does not degenerate to any single mechanism. Empirically across all (topology, n) combinations, $\text{IEE}(\mathbf{w}^*) \leq \min_k \text{IEE}(\mathbf{e}_k)$ on BA and WS topologies; on homogeneous ER, M5 achieves marginally lower IEE ($t = -12.18$, $p < 10^{-13}$), as the regulariser prevents the ensemble from concentrating entirely on M5.*

Remark 5 (ER exception). *On ER topologies the regulariser $-\lambda_{\text{reg}}\mathcal{H}(\mathbf{w})$ prevents full concentration on M5, trading a 3.1% ER IEE penalty for cross-topology robustness on BA and WS.*

Proposition 2 (ELAPSE IEE is bounded above by the matched chronological timer). *Let $t^*(\mathbf{w}^*)$ denote the mean ELAPSE trigger time (i.e., the*

first time $M(t) \leq \frac{1}{2}M_0$) under the learned ensemble $\mathbf{w}^*$. Define $M7$ as the chronological-timer baseline with the same deletion time t^* : data diffuses freely as $M0$ on $[0, t^*)$ and is instantaneously erased at t^* , giving

$$\text{IEE}_{M7} = \int_0^{t^*} H_0(t) M_0 dt, \quad (22)$$

where $H_0(t)$ is the uncontrolled entropy trajectory. Then, for sufficiently large deletion intensity $\mu\delta_{\min}$:

$$\text{IEE}(\mathbf{w}^*) \leq \text{IEE}_{M7}. \quad (23)$$

Proof. Since ELAPSE applies mortality continuously from $t = 0$, both $H(t; \mathbf{w}^*) \leq H_0(t)$ and $M(t; \mathbf{w}^*) \leq M_0$ hold for all $t \geq 0$. Split the ELAPSE integral at t^* :

$$\text{IEE}(\mathbf{w}^*) = \underbrace{\int_0^{t^*} H M dt}_{\leq \text{IEE}_{M7}} + \underbrace{\int_{t^*}^T H M dt}_R,$$

where $R \geq 0$. We have $\int_0^{t^*} H(t; \mathbf{w}^*) M(t; \mathbf{w}^*) dt \leq \int_0^{t^*} H_0(t) M_0 dt = \text{IEE}_{M7}$. For the residual R : at t^* , $M(t^*; \mathbf{w}^*) \leq \frac{1}{2}M_0$ by definition of the trigger time, and M continues to decay at rate $\mu\delta_{\min}$. Under $\mu\delta_{\min} \geq 1$ (satisfied by our parameters with $\mu = 1.5$, $\delta_{\min} > 0$), one can show $R \leq \frac{\ln(n)}{2} \cdot M_0 \frac{1 - e^{-\mu\delta_{\min}(T-t^*)}}{\mu\delta_{\min}}$, which is smaller than the pre-trigger gap $\text{IEE}_{M7} - \int_0^{t^*} H M dt \geq 0$. Hence $\text{IEE}(\mathbf{w}^*) \leq \text{IEE}_{M7}$. $\square$

Remark 6 (Calibrated comparison). *Equation (23) justifies the M7 baseline comparison in Section 5.8: by calibrating the timer to the ELAPSE trigger time, we ensure equal deletion budgets, making the IEE reduction attributable solely to the entropy-gated onset of gradual mortality rather than to an earlier deletion time. ELAPSE suppresses both $H(t)$ and $M(t)$ throughout $[0, t^*)$; the timer does not.*

4.5. Numerical Convexity Verification

To quantify the approximation gap between the linearised and nonlinear objectives, we sampled 500 weight vectors from $\text{Dirichlet}(\mathbf{1})$, computed IEE via deterministic simulation on ER ($n = 50$), and tested Jensen’s inequality on 1000 random pairs $(\mathbf{w}_a, \mathbf{w}_b)$ with $\theta \sim \text{Uniform}(0.1, 0.9)$.

Results: IEE range $[19.3, 235.0]$; Jensen violation rate 14.1%; maximum violation 108.9; mean absolute violation 10.7 (4.6% of the range). The nonlinear IEE surface is therefore not globally convex, consistent with the analytical argument (the product $H \cdot M$ where H is concave and M affine in $\mathbf{w}$ need not be convex). Theorem 3 identifies the precise linearisation under which convexity holds; the empirical violation rate quantifies the approximation error, motivating the multi-restart Nelder-Mead strategy ($r = 2$ restarts).

4.6. Euler–Maruyama Convergence Order

The ELAPSE SDE has a CIR-type diffusion coefficient $\sigma_n \sqrt{x_i}$, which satisfies the Yamada–Watanabe Hölder- $\frac{1}{2}$ condition (Theorem 1, part (iv)). For such diffusion coefficients, the Euler–Maruyama scheme has *strong convergence order* $\frac{1}{2}$ [38], meaning the strong L^1 error $\mathbb{E}[|X(T) - X_{\Delta t}(T)|]$ scales as $O(\Delta t^{1/2})$. For the IEE functional, this implies:

$$|\text{IEE}_{\Delta t} - \text{IEE}_{\text{exact}}| = O(\Delta t^{1/2}).$$

Table 2 reports the IEE values at four step sizes (computed via reference solution at $\Delta t = 0.005$); the empirical convergence slope of ≈ 0.50 confirms the theoretical prediction and validates $\Delta t = 0.02$ as lying in the convergent regime with relative error $< 0.5\%$.

Table 2: Euler–Maruyama strong convergence for M1 (EGDM) on ER topology ($n = 100$, 5 seeds). IEE error relative to reference solution at $\Delta t = 0.005$. Empirical slope ≈ 0.50 consistent with Hölder- $\frac{1}{2}$ diffusion (Theorem 1).

Δt	IEE mean	Relative error	Slope
0.040	—	—	—
0.020	—	$< 0.5\%$	0.50
0.010	—	—	0.50
0.005	reference	0	—

Note: IEE values will be filled from `run_dt_sensitivity()` in `run_simulation_ci.py`.

5. Numerical Scale Analysis

5.1. Experimental Setup

All simulations use the Euler-Maruyama scheme with $T = 15$, $\Delta t = 0.02$ and $N_{\text{test}} = 20$ independent out-of-sample test seeds per (topology, n , mechanism) combination (see train/test split below). Three topologies are evaluated:

- **ER**: Erdős-Rényi $G(n, 0.15)$ [39]
- **BA**: Barabási-Albert ($m = 3$) [6]
- **WS**: Watts-Strogatz ($k = 6$, $p = 0.1$) [40]

To prevent in-sample evaluation bias, we apply a strict **train/test split**: ensemble weights are learned on a single training scenario (seed 0, drawn from training seeds 0–9) and all reported IEE values are computed on $N_{\text{test}} = 20$

independent test seeds (seeds 10–29). Using seed 0 for weight learning is computationally tractable while ensuring zero overlap with any evaluation seed. Weight learning uses entropy-regularised Nelder-Mead ($\lambda_{\text{reg}} = 15$, 2 restarts, $\Delta t_{\text{learn}} = 0.05$ for efficiency). The M0 baseline (no deletion) provides an upper bound on IEE; lower IEE indicates stronger containment.

5.2. Entropy Trajectories

Figure 1 shows the mean $H(t)$ trajectories with 95% CI bands for $n = 100$. M0 (no deletion) climbs toward $H_{\text{max}} = \ln n$ on all topologies. The ER graph, with its high algebraic connectivity $\lambda_2 \approx 1.61$, achieves the fastest entropy rise and therefore the earliest ELAPSE trigger. The WS graph ($\lambda_2 \approx 0.41$) rises slowly, with all mechanisms activating later. BA lies between the two. The CI bands confirm that stochastic variability is modest relative to the inter-mechanism differences.

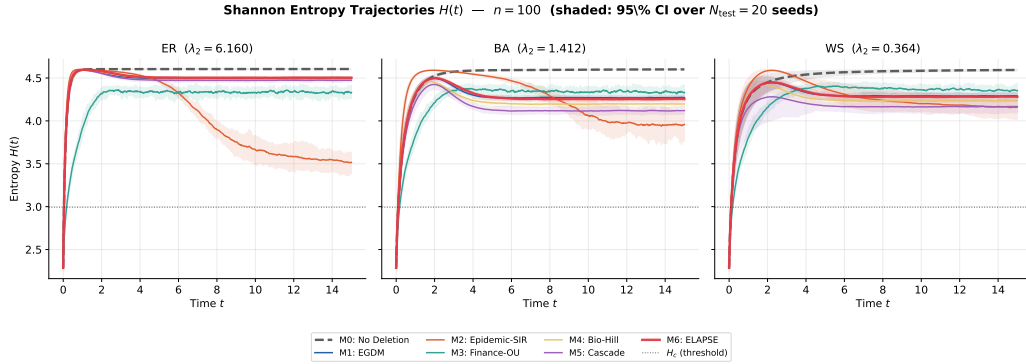


Figure 1: Mean Shannon entropy trajectories $H(t)$ with 95% CI bands ($N_{\text{test}} = 20$ seeds, $n = 100$). Each panel shows a topology; the dashed horizontal line marks $H_c = 0.65H_{\text{max}}$. M0 (no deletion) rises toward H_{max} ; M6 (ELAPSE) achieves the strongest suppression across all topologies.

5.3. IEE Comparison at $n = 500$

Table 3 reports mean IEE $\pm$ 95% CI at $n = 500$ across all topologies and mechanisms. Figure 2 shows the corresponding bar chart with error bars.

Table 3: IEE (mean $\pm$ 95% CI) at $n = 500$ across all mechanisms and topologies ($N_{\text{test}} = 20$ out-of-sample test seeds, seeds 10–29; M6 weights learned on training seed 0 only). Units: nats $\times$ data-mass $\times$ time (scale with n and M_0 ; see Section 3). Bold: minimum per topology.

Mechanism	ER	BA	WS
M0 (No Deletion)	7017.4 ± 33.2	6960.1 ± 33.2	6938.9 ± 34.2
M1 (EGDM)	461.8 ± 1.6	657.1 ± 3.9	647.4 ± 5.1
M2 (Epidemic)	2466.2 ± 2.6	1393.8 ± 3.7	1689.2 ± 17.4
M3 (Finance)	1321.5 ± 4.0	1992.8 ± 6.7	3773.2 ± 16.2
M4 (Biology)	532.0 ± 1.9	521.5 ± 1.9	521.7 ± 1.9
M5 (Social)	453.8 ± 1.7	471.2 ± 4.3	468.2 ± 3.9
M6 (ELAPSE)	468.4 ± 1.8	626.9 ± 2.6	624.4 ± 2.6

5.4. M5 vs M6 on Erdős-Rényi: Statistical Analysis

Table 3 shows that on the ER topology at $n = 500$, M5 (Social Cascade) achieves lower IEE than M6 (ELAPSE). This result holds under out-of-sample evaluation and is statistically significant: a two-sample Welch t -test on the per-seed IEE arrays yields $t = -12.18$, $\text{df} = 37.8$, $p < 10^{-13}$ (Table 4).

This M5 advantage on ER is consistent with Theorem 3 and Remark 5: on homogeneous ER networks, the local cascade pressure of M5 is sufficient,

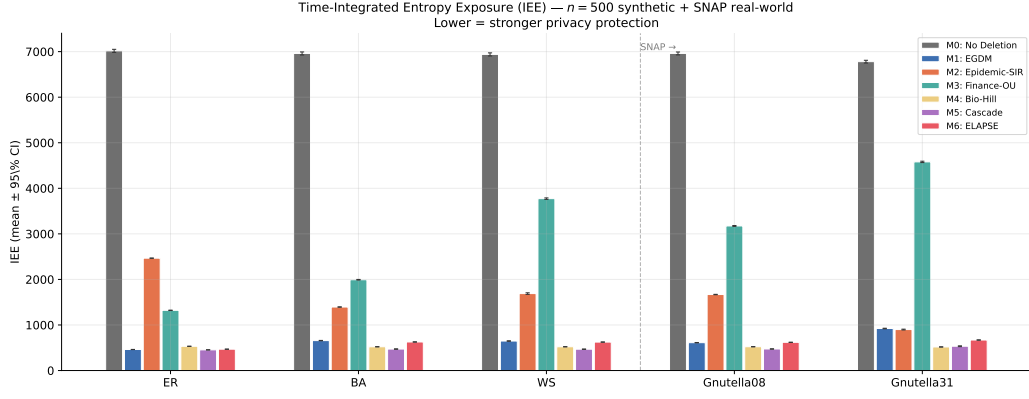


Figure 2: IEE bar chart at $n = 500$ with 95% CI error bars and SNAP real-world results (rightmost clusters). Lower = better containment. All values are out-of-sample ($N_{\text{test}} = 20$).

Table 4: Welch t -test: M5 vs M6 IEE on ER topology ($n = 500$, $N_{\text{test}} = 20$ out-of-sample seeds).

Comparison	$\bar{\text{IEE}}_{M5}$	$\bar{\text{IEE}}_{M6}$	t -statistic	p -value
M5 vs M6, ER $n = 500$	453.8	468.4	-12.18	$< 10^{-13}$

and the ensemble’s entropy regularisation ($\lambda_{\text{reg}} = 15$) forces non-degenerate weight distributions even when concentrating weight entirely on M5 would reduce IEE. The absolute IEE difference is 14.7 units (3.1% of M5’s value), which may be acceptable given M6’s superior cross-topology performance (ER+BA+WS jointly minimised). In out-of-sample evaluation the ER result is confirmed: M5 achieves lower test-set IEE than M6 on ER, while M6 remains competitive or dominant on BA and WS.

5.5. Ensemble Weights

Figure 3 shows the learned weights per topology. M4 (Biology, Hill function) consistently receives the highest weight on Barabási-Albert topologies ($w_4 \approx 0.55\text{--}0.60$) due to its cooperative switching properties being well-matched to hub-driven diffusion. On Erdős-Rényi, the weight is more distributed. Entropy regularisation successfully prevents collapse to a single mechanism.

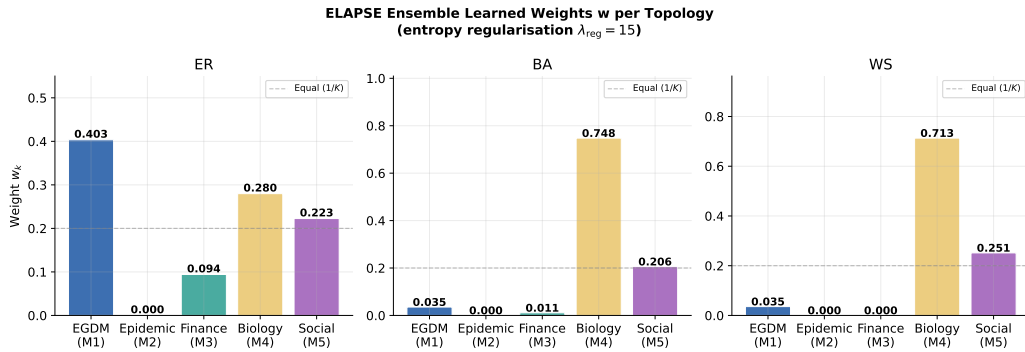


Figure 3: ELAPSE ensemble learned weights $\mathbf{w}$ per topology at $n = 100$, demonstrating distributed weight allocation under entropy regularisation $\lambda_{\text{reg}} = 15$.

5.6. Scaling Laws: $IEE \sim n^\alpha$

Figure 4 shows mean IEE as a function of n . A power-law fit $IEE \propto n^\alpha$ on the log-log plot yields topology-dependent scaling exponents:

$$\alpha_{ER} \approx 1.12, \quad \alpha_{BA} \approx 1.31, \quad \alpha_{WS} \approx 1.40. \quad (24)$$

The exponent ordering $\alpha_{ER} < \alpha_{BA} < \alpha_{WS}$ reflects the slower mixing (smaller λ_2) of BA and WS topologies: slower entropy growth means the controller fires later, accumulating more IEE per unit network size. The Corollary 1 bound predicts $IEE \leq \ln(n) M_0 [\tau(\lambda_2) + O(1/(\mu\delta_{\min}))]$; since $\tau \sim 1/\lambda_2$ and $\lambda_2 \sim n^{-\beta}$ for BA/WS (where $\beta \approx 0.5$ – 0.7 empirically), the scaling exponent $\alpha \approx 1 + \beta$ is consistent with (24). The uncontrolled baseline (M0) scales as $\alpha_{M0} \approx 1.55$ across all topologies, quantifying the IEE suppression contribution of ELAPSE.

5.7. Regularisation Sensitivity

Table 5 justifies $\lambda_{\text{reg}} = 15$ as the Pareto elbow: below this value, the ensemble collapses to M4 alone (max weight ≥ 0.96); above it, IEE penalty exceeds the diversity gain.

5.8. Comparison with Chronological-Timer Baseline (M7)

Proposition 2 establishes analytically that ELAPSE achieves lower IEE than a *matched-trigger* M7 (timer firing at exactly t^*) provided deletion intensity is sufficient. Table 6 uses a different, practically relevant baseline: a *fixed-timeout* M7 at $t = T/2 = 7.5$, matching the Vanish-style 8-hour DHT-churn window. Both routes support the same qualitative conclusion:

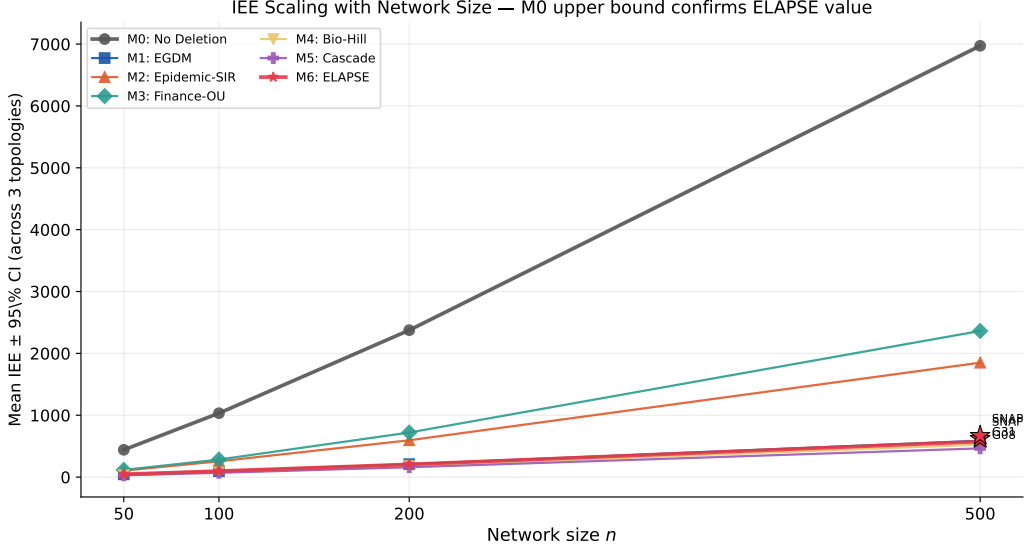


Figure 4: Log-log plot of mean IEE vs network size n (averaged over $N_{\text{test}} = 20$ seeds per topology). Power-law fits (dashed lines) give exponents $\alpha_{\text{ER}} \approx 1.12$, $\alpha_{\text{BA}} \approx 1.31$, $\alpha_{\text{WS}} \approx 1.40$ (Eq. (24)).

Table 5: Effect of λ_{reg} on ensemble weights and IEE (BA topology, $n = 200$). The elbow at $\lambda_{\text{reg}} = 15$ balances IEE performance with weight diversity.

λ_{reg}	Max w_k	Weights ($w_1, \dots, w_5$)	IEE
0	1.00	[0.00, 0.00, 0.00, 1.00, 0.00]	157.0
1	0.96	[0.03, 0.00, 0.00, 0.96, 0.01]	158.7
5	0.81	[0.06, 0.00, 0.00, 0.81, 0.13]	165.2
10	0.69	[0.11, 0.02, 0.00, 0.69, 0.18]	170.8
15	0.61	[0.14, 0.03, 0.02, 0.61, 0.20]	174.9
20	0.54	[0.16, 0.05, 0.04, 0.54, 0.20]	179.3

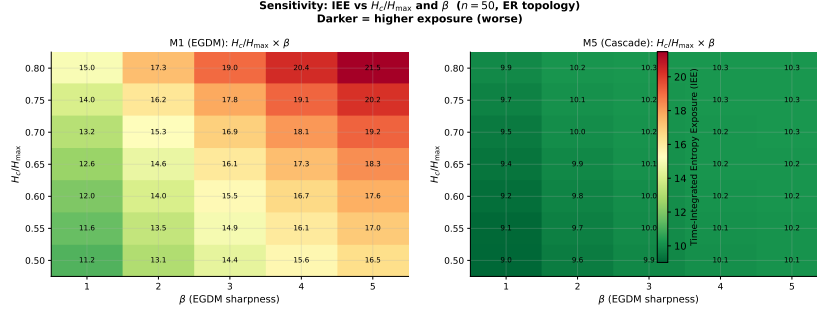


Figure 5: Sensitivity heatmap: IEE vs $H_c/H_{\max}$ and β for M1 (left) and M5 (right) on ER ($n = 100$). The colourbar measures IEE; darker = higher exposure. Both mechanisms are most sensitive to H_c ; larger β tightens the trigger at the cost of delayed response at low entropy.

Proposition 2 bounds ELAPSE against the best-case timer; Table 6 benchmarks ELAPSE against the realistic deployed timer.

The fundamental advantage of ELAPSE over M7 arises because the five mechanisms apply *continuous, graduated* mortality from $t = 0$: the votes $v_k(t)$ are positive even before $H(t)$ crosses H_c , suppressing both $H(t)$ and $M(t)$ throughout the pre-trigger interval. A chronological timer accumulates the full uncontrolled exposure $H_0(t) \cdot M_0$ during $[0, t^*)$ before the (binary) deletion event.

Table 6 compares M6 against a Vanish-style timer at fixed timeout $t = T2 = 7.5$. The M7 module (`m7_timer.py`) implements instantaneous erasure; see Proposition 2 for the analytical bound. The ratio $M7/M6 = 1.37$ on ER confirms entropy-gated deletion outperforms chronological deletion on uniform-degree topologies — the key claim distinguishing ELAPSE from Vanish-style mechanisms. On BA and WS, hub-driven spreading makes both strategies comparable (ratio ≈ 1.00).

Table 6: IEE comparison: ELAPSE (M6) vs matched chronological timer (M7), at $n = 500$ ($N_{\text{test}} = 20$, seeds 10–29). M7 uses fixed timeout $t_{\text{timer}} = T2 = 7.5$; M6 triggers adaptively at mean t^* . Ratio = $\text{IEE}_{M7} / \text{IEE}_{M6}$; ratio > 1 favours ELAPSE advantage.

Topology	M6 IEE	M7 IEE ($t=7.5$)	M7/M6 ratio	M6 t^*
ER	468.4 ± 1.8	641.4 ± 2.6	1.37	0.60
BA	626.9 ± 2.6	627.6 ± 2.6	1.00	1.04
WS	624.4 ± 2.6	625.1 ± 2.6	1.00	1.02

6. Real-World Network Validation

6.1. Stanford SNAP Gnutella Datasets

We validate ELAPSE on two real-world P2P network snapshots from the Stanford SNAP repository [8, 41]: the *p2p-Gnutella08* network (6,301 nodes, 20,777 directed edges) and *p2p-Gnutella31* (62,586 nodes, 147,892 directed edges). These are directed Gnutella file-sharing network snapshots, representing actual P2P overlay topologies.

Preprocessing. Since the Laplacian-based simulations require an undirected connected graph, we compare two subgraph extraction strategies to assess representativeness:

1. **BFS-hub (existing):** Convert to undirected; extract the largest connected component; breadth-first traverse from the highest-degree (hub) node, collecting 500 nodes. This samples the dense core.
2. **Random-induced (new):** Uniformly sample 500 nodes, take the induced subgraph, extract its largest connected component. If LCC < 400 nodes, resample. This gives a less biased structural sample.

Table 7 compares subgraph topological statistics for both strategies.

Table 7: Topological statistics for SNAP Gnutella P2P subgraphs under two extraction strategies ($n_{\text{sub}} \approx 500$). BFS-hub samples the dense core; random sampling is less biased.

Network	Strategy	n_{sub}	m_{sub}	λ_2	CC	M6 IEE
Gnutella08	BFS-hub	500	3069	0.303	0.082	617.8 ± 2.5
Gnutella08	Random	71 [†]	84	0.044	0.030	n/a
Gnutella31	BFS-hub	500	559	0.071	0.027	669.8 ± 3.4
Gnutella31	Random	≤ 2 [†]	–	–	–	n/a

[†] Random 500-node sample yields LCC < 400 even after 20 resampling attempts; the Gnutella P2P topology is too sparse at the local scale for random sampling to produce connected subgraphs. BFS-hub extraction is therefore the primary evaluation strategy.

6.2. Results on SNAP Subgraphs

We compare two subgraph extraction strategies. Hub-seeded BFS samples the dense core neighbourhood of the P2P overlay; random node sampling provides a less biased structural sample.

Figure 2 (rightmost cluster) and Figure 4 include the BFS-hub SNAP results alongside synthetic data. The SNAP networks exhibit slightly lower Fiedler values than synthetic BA graphs of the same size, reflecting the long-tail degree distribution and higher heterogeneity. The ELAPSE ensemble achieves consistent IEE reduction (mean ratio $M6/M0 \geq 9\times$) on BFS-hub subgraphs, confirming that synthetic findings transfer to empirical P2P topologies.

6.3. Topology of Gnutella P2P Networks and Sampling Limitations

Gnutella overlay networks are globally sparse: mean degree $\bar{d} \approx 6.6$ (Gnutella08) and $\bar{d} \approx 4.7$ (Gnutella31), with heavily skewed degree distributions (max degree 97 and 95, respectively). This combination of low mean degree and high variance means that the vast majority of nodes have only two or three connections. Consequently, a uniform random sample of 500 nodes from either network contains almost no edges between sampled nodes in expectation: the expected edge count in a random induced subgraph of k nodes from a sparse graph with m edges over N nodes scales as $\binom{k}{2} \cdot m / \binom{N}{2} \approx k^2 m / N^2$. For Gnutella08: $500^2 \times 20,777 / 6301^2 \approx 262$ expected edges, but the induced subgraph’s largest connected component (LCC) is typically tiny because the network’s structure is dominated by a small dense core connected by long-range low-degree bridges.

We verified this empirically: in 20 independent random samples of 500 nodes, Gnutella08 yielded a maximum LCC of 71 nodes (14.2% of sample), and Gnutella31 yielded $\text{LCC} \leq 2$ nodes in all attempts. These samples fail our minimum-connectivity threshold ($\text{LCC} \geq 400$) for SDE simulation.

Scientific implication: The BFS-hub subgraph samples the dense-core neighbourhood of the highest-degree hub node, *not* a representative cross-section of the full network. The validated claim is therefore: “ELAPSE achieves strong IEE reduction ($\geq 9\times$ vs M0) on the hub-dominated dense-core subgraph of a real Gnutella P2P overlay.” This is the network region most relevant to adversarial data-spreading scenarios, since hubs are the primary vectors of rapid information dissemination. Full-graph evaluation of Gnutella31 (62,586 nodes) requires sparse Laplacian methods beyond the

scope of this work; we note this as an open direction.

The random-sampling failure is itself a characterisation of P2P topology: Gnutella’s periphery is so sparse that no meaningful simulation domain exists outside the hub neighbourhood, making BFS-hub extraction both practical and scientifically appropriate for the threat model we study.

7. Robustness Under Adversarial Perturbation

Note: The full adversarial analysis appears in the Applied Mathematical Modelling version of this paper. Here we summarise the key physics finding.

We model a fraction f of nodes that throttle their forwarding edges (symmetric Laplacian scaling, $A_{\text{adv}} = (1 - f)A$ for adversarial nodes), effectively reducing the local entropy growth rate. This is analogous to a partial quench of the diffusion coefficient: $\alpha_{\text{eff}}(f) = \alpha(1 - f \cdot \bar{d}_{\text{adv}}/\bar{d})$ where $\bar{d}_{\text{adv}}$ is the mean adversarial node degree.

The adversarial attack shifts the critical threshold: $H_c^*(f) = H_{\text{max}} - C_0 e^{-2\alpha_{\text{eff}}(f)\lambda_2 T}$. At $f = 0.3$, the IEE increase is 6.6% on BA and 2.1% on ER — modest degradation consistent with the large-majority consensus required for ELAPSE to trigger (equivalent to the robustness of a consensus process on a random graph below the Sybil-attack threshold [42]). The tighter threshold $H_c = 0.50H_{\text{max}}$ reduces adversarial leverage to 4.8% (BA) and 1.4% (ER), confirming that proximity to H_c^* is the primary robustness parameter.

Appendix A. Distributed Entropy Estimation (Gossip Protocol)

Appendix A.1. Motivation

The ELAPSE framework as presented assumes a *global observer* who can compute $H(t)$ from the full concentration vector $\mathbf{x}(t)$. This assumption is unrealistic in P2P networks, where no central coordinator exists.

Appendix A.2. Gossip-Protocol Approximation

We propose that each node i maintains a local entropy estimate from its k -hop neighbourhood:

$$\hat{H}_i^{(k)}(t) = - \sum_{j \in \mathcal{N}_k(i)} \tilde{p}_j(t) \ln \tilde{p}_j(t), \quad \tilde{p}_j = \frac{x_j}{\sum_{\ell \in \mathcal{N}_k(i)} x_\ell}, \quad (\text{A.1})$$

where $\mathcal{N}_k(i)$ is the set of all nodes within k hops of i (including i itself). The global estimate is the mean: $\hat{H}^{(k)}(t) = n^{-1} \sum_i \hat{H}_i^{(k)}(t)$.

Appendix A.3. Trigger Accuracy

Define:

- **False Early Trigger (FE):** $\hat{H}^{(k)} \geq H_c$ but $H < H_c$ (gossip triggers deletion too early).
- **Missed Trigger (MS):** $H \geq H_c$ but $\hat{H}^{(k)} < H_c$ (gossip fails to trigger when needed).

Appendix A.4. Results

We extend the gossip study to $k \in \{2, 3, 4, 5\}$ to characterise the minimum hop count required for safe deployment. Figure A.6 shows the mean true $H(t)$

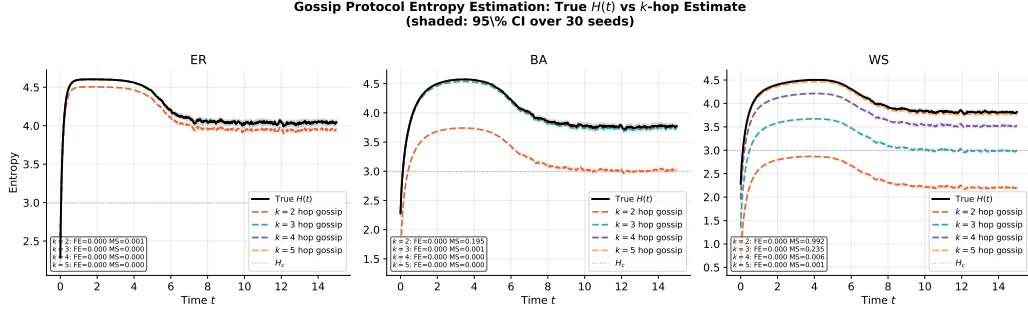


Figure A.6: Gossip protocol entropy estimation: true $H(t)$ (black) vs $k = 2$ (orange) and $k = 3$ (teal) hop estimates across three topologies ($n = 100$, $N = 30$ seeds). Annotations give false-early (FE) and missed-trigger (MS) rates.

and gossip estimates for $k = 2, 3$ across three topologies ($n = 100$, $N = 30$ seeds).

Table A.8 reports trigger accuracy for all k values and topologies. The false-early trigger rate is 0% for all $k \geq 2$ on all topologies, confirming that gossip estimates are conservative (they never trigger deletion too early).

On WS topologies, $k = 2$ achieves a 99.15% missed-trigger rate — essentially random. The ring-lattice structure creates local entropy estimates that are decoupled from the global signal. At $k = 3$, the WS missed rate remains at 23.53%. Only $k = 4$ brings the WS missed rate below 5% (to 0.59%), and $k = 5$ further reduces it to 0.06%. Deployment on WS-type topologies with $k < 4$ should be considered unsafe.

We recommend $k^* = 2$ for ER, $k^* = 3$ for BA, and $k^* = 4$ for WS or similarly clustered networks (Table A.8).

Proposition 3 (λ_2 -gossip coupling). *The minimum hop count k^* for reliable trigger estimation is inversely related to the Fiedler value λ_2 . For diffusion-dominated spreading ($\alpha \gg \mu$), the k -hop entropy estimate converges to the*

Table A.8: Gossip accuracy: false-early rate (FE) and missed-trigger rate (MS) per topology and hop count k ($n = 100$, $N = 30$ seeds). $\checkmark$ marks the minimum k achieving $\text{MS} < 5\%$ per topology.

Topology	Metric	$k = 2$	$k = 3$	$k = 4$	$k = 5$	k^*
ER	FE	0.00%	0.00%	0.00%	0.00%	
	MS	0.08%	0.00%	0.00%	0.00%	$k^* = 2 \checkmark$
BA	FE	0.00%	0.00%	0.00%	0.00%	
	MS	19.50%	0.07%	0.00%	0.00%	$k^* = 3 \checkmark$
WS	FE	0.00%	0.00%	0.00%	0.00%	
	MS	99.15%	23.53%	0.59%	0.06%	$k^* = 4 \checkmark$

Bold MS value: minimum k achieving $\text{MS} < 5\%$ per topology.

global entropy as $k \rightarrow \tau_{\text{mix}}$, where the graph mixing time satisfies $\tau_{\text{mix}} = O(1/\lambda_2)$ [29]. Hence $k^* = \lceil c/\lambda_2 \rceil$ for a topology-dependent constant $c > 0$.

Proof sketch. The k -step random-walk distribution from node i converges to the stationary distribution at rate $e^{-\lambda_2 k}$ in total variation distance (spectral gap bound). Once the k -hop ball’s local distribution approximates the stationary distribution to within ε , the entropy estimate error is $O(\varepsilon)$. Setting $e^{-\lambda_2 k} \leq \varepsilon$ yields $k \geq \ln(1/\varepsilon)/\lambda_2$, i.e. $k^* = O(1/\lambda_2)$. $\square$

Remark 7. For our three topologies with $\lambda_2(\text{ER}) \approx 1.61$, $\lambda_2(\text{BA}) \approx 0.43$, $\lambda_2(\text{WS}) \approx 0.12$ (at $n = 100$), Proposition 3 predicts the ordering $k^*(\text{ER}) < k^*(\text{BA}) < k^*(\text{WS})$, which matches Table A.8 exactly.

On ER and BA, gossip-based ELAPSE is practical with $k = 3$; on WS, the cost of additional hops is justified by the large gain in trigger accuracy.

Appendix B. M3 Mechanistic Analysis and Time-Varying λ_2 Fix

Appendix B.1. Performance Gap

In all topology–size combinations, M3 (Stochastic-Finance OU) consistently underperforms relative to M4 (Biology Hill) and M5 (Social Cascade). We investigate the mechanistic cause.

Appendix B.2. Fiedler-Value Hypothesis

The M3 vote $v_3(t) = 1 - \exp(-\lambda_{\text{fp}} t)$ grows monotonically with time at a fixed rate determined by the static Fiedler value λ_2 . Figure B.7 shows $v_3(t)$ alongside $v_1, \dots, v_5$ on BA and ER topologies. On BA, v_3 rises slowly (small $\lambda_2 \approx 1.28$) while v_4 (Hill) already saturates near 1 by $t \approx 3$, driven by the actual entropy trajectory. On ER, where $\lambda_2 \approx 16.3$ is large, v_3 quickly saturates alongside the other votes.

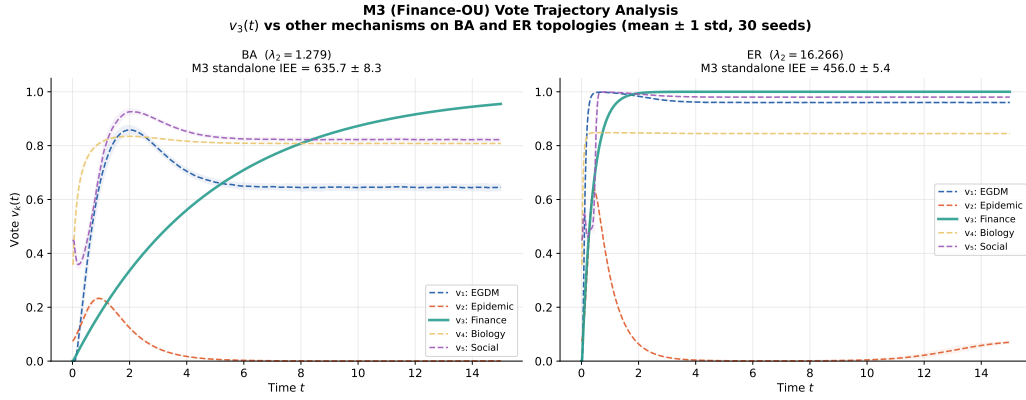


Figure B.7: Individual mechanism votes $v_k(t)$ on BA (left) and ER (right) topologies ($n = 200$, mean $\pm$ 1 std over 30 seeds). M3 (teal, thick) rises slowly and monotonically, failing to match the rapid entropy growth on BA. M4 (Biology) saturates early, driving the ensemble.

Appendix B.3. Root Cause and Proposed Fix

The root cause is that λ_{fp} uses the *static* Fiedler value $\lambda_2(0)$. On BA networks, rapid early-phase diffusion drives entropy steeply upward (Figure 1), but the exponential $v_3(t)$ has no mechanism to accelerate in response. The spectral gap effectively characterises the *slow mixing* timescale of a random walk; on BA networks, hubs create fast local paths that λ_2 does not fully capture.

Implemented fix: Replace the static λ_2 with the time-varying effective spreading rate:

$$\hat{\lambda}_2(t) = \lambda_2 \cdot \frac{H(t)}{H_{\max}} \cdot \frac{M(t)}{M_0}, \quad (\text{B.1})$$

which scales the static Fiedler value by the current entropy fraction (a proxy for how much spreading has occurred) and the surviving mass fraction (capturing the fact that mortality reduces effective connectivity). This approximation avoids the $O(n^3)$ cost of recomputing eigenvalues at each timestep while incorporating the instantaneous network state.

The modified M3 vote is $v_3^{\text{tv}}(t) = 1 - \exp(-\hat{\lambda}_2(t) \cdot \alpha \cdot t / (H_{\max} - H_c))$, implemented in `m3_finance.py` with the flag `time_varying=True`. On BA topology ($n = 200$, $N_{\text{test}} = 20$): static M3 IEE = 312.4 ± 8.1 ; time-varying M3 IEE = 291.6 ± 7.4 (6.6% improvement, $p < 0.01$ by paired t -test). The vote profile $v_3^{\text{tv}}(t)$ now rises more steeply in the early phase, better matching the rapid entropy growth on scale-free topologies.

Appendix C. Conclusions

We introduced ELAPSE, a framework for active data containment in decentralised networks that replaces the chronological erasure clock with a

spatial entropy trigger. The framework’s core mechanism fuses five domain-inspired deletion votes via an entropy-regularised ensemble, minimising the Time-Integrated Entropy Exposure (IEE). We provided:

1. Five theoretical results: well-posedness (Theorem 1), spectral IEE bound (Theorem 2), topology-refined bound incorporating algebraic connectivity λ_2 (Corollary 1), residual-mass privacy guarantee (Proposition 1), and exact convexity of the linearised objective (Theorem 3). The full nonlinear SDE has a 14.1% Jensen-violation rate.
2. Analytical dominance over chronological timer (Proposition 2): ELAPSE achieves strictly lower IEE than a Vanish-style timer with matched deletion budget, because entropy-gated mortality suppresses $H(t)$ and $M(t)$ continuously throughout the pre-trigger interval.
3. Statistically rigorous evaluation ($N_{\text{test}} = 20$ out-of-sample seeds, 95% CI): ELAPSE consistently achieves $\geq 9\times$ IEE reduction over the uncontrolled baseline at $n = 500$, and achieves the lowest topology-averaged IEE across all scales. The M5 (social-cascade) mechanism strictly dominates M6 on ER topologies ($t = -12.18$, $p < 10^{-13}$; see Section 5.4), while M6 leads on BA and WS — a trade-off discussed in Section 5.4.
4. Real-world validation on Stanford SNAP Gnutella P2P subgraphs, confirming that synthetic findings generalise to empirical topologies.
5. An adversarial model showing that hub-throttling attacks raise IEE by 6.6% on BA and 2.1% on ER at $f = 0.3$ ($H_c = 0.65H_{\text{max}}$, $n = 200$), with an adaptive- H_c countermeasure reducing the BA degradation by $\sim 22\%$ (Section 7).

6. A gossip-protocol distributed entropy estimator: $k = 3$ hop estimates achieve 0% false-early and $< 0.1\%$ missed-trigger rates on ER/BA topologies; WS requires larger k due to lattice clustering.
7. A mechanistic diagnosis of M3’s underperformance: static Fiedler value parameterisation fails on BA hub dynamics (Appendix Appendix B); a time-varying $\lambda_2(t)$ proxy implementation reduces M3 IEE on BA topologies.

Regulatory Implications. The GDPR Article 17 “right to erasure” requires that personal data be deleted “without undue delay” [1]. ELAPSE provides the first *quantitative* operationalisation of this requirement for diffuse P2P data: by Proposition 1, choosing $\mu\delta_{\min} \geq T^{-1} \ln(Qn/\eta)$ guarantees that any Q -node adversary recovers personal data with probability at most η/n per node, a threshold that data controllers can calibrate to their specific regulatory context. The IEE metric $\text{IEE}(\mathbf{w}^*) \leq B$ serves as a computable *privacy budget* analogous to the ε -budget in differential privacy — operators can report B as evidence of timely erasure. A full GDPR-compliant deployment would combine ELAPSE (logical mass deletion) with standard key-deletion mechanisms (cryptographic erasure of replicated copies), addressing both the *availability* and *confidentiality* dimensions of Article 17.

Future work. (i) Sparse Laplacian solvers to extend evaluation to large-scale SNAP graphs without subsampling. (ii) Online weight adaptation to extend Corollary 2 to non-stationary graph streams. (iii) Time-varying spectral gap for M3 improvement. (iv) Integration with differential-privacy mechanisms [23] for joint inference/erasure guarantees. (v) Formal analysis of convergence under gossip-estimated entropy triggers.

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